

The Elastic Vacuum: A Sequestering Mechanism for Vacuum Energy and the Complete Dynamical History of Cosmic Expansion

Boris Kriger^{1,2}

¹ Information Physics Institute, Gosport, Hampshire, United Kingdom

`boris.kriger@informationphysicsinstitute.net`

² Institute of Integrative and Interdisciplinary Research, Toronto, Canada

`boriskruger@interdisciplinary-institute.org`

ORCID: [0009-0001-0034-2903](https://orcid.org/0009-0001-0034-2903)

Abstract

The cosmological constant problem—the 10^{55} – 10^{120} order discrepancy between quantum field theory estimates of vacuum energy and its observed cosmological value—rests on the 1967 Zel’dovich identification of vacuum energy with the cosmological constant. This paper develops a dynamical sequestering mechanism in which vacuum energy gravitates locally within bound structures but is screened from cosmological expansion. We introduce a physical analogy—the elastic membrane model—and show that it is not merely pedagogical: the Einstein–Hilbert action is an elastic energy functional, gravitational waves confirmed by LIGO are elastic oscillations, and Sakharov’s induced gravity programme derives the elastic modulus from quantum field theory. Spacetime is, in a physically precise sense, an elastic medium. This grounding yields a complete dynamical history: from an initially highly curved state (Big Bang), through mutual relaxation of matter and spacetime geometry, to the present epoch of accelerating expansion driven by the membrane’s residual elasticity. The model yields three key results. First, we show that the sequestering mechanism preserves the equivalence principle locally while modifying global vacuum contributions to the Friedmann equations. Second, we demonstrate that the observed near-flatness of the universe need not be an initial condition requiring inflation, but may be the natural late-time outcome of geometric relaxation—with a positive feedback loop in which decreasing curvature facilitates faster expansion. Third, we identify the transition from deceleration to acceleration at $z \approx 0.7$ as the

epoch at which geometric elasticity begins to dominate over matter-driven dynamics. The framework uses only established physics—general relativity, quantum field theory, and $E = mc^2$ —without invoking new particles, new forces, or fine-tuning.

Keywords: vacuum energy, cosmological constant, sequestering mechanism, dark energy, dark matter, flatness problem, elastic vacuum, cosmic expansion, Zel'dovich identification

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1 Introduction

The cosmological constant problem remains one of the most persistent puzzles in modern theoretical physics. Quantum field theory predicts an enormous vacuum energy density, while cosmological observations indicate that the effective vacuum energy influencing the expansion of the universe is extremely small. In the standard Λ CDM model this tension is handled by introducing the cosmological constant Λ as a geometric parameter of spacetime, assumed to be uniform and constant. However, this approach leaves open a fundamental conceptual question: if vacuum energy is a real physical property of quantum fields, why does it not dominate cosmic expansion?

The standard account traces the problem to Zel’dovich [Zel’dovich, 1967, 1968], who observed that the vacuum stress-energy tensor takes the Lorentz-invariant form $\langle T_{\mu\nu} \rangle_{\text{vac}} = -\rho_{\text{vac}} g_{\mu\nu}$, mathematically identical to a cosmological constant with $\Lambda = 8\pi G \rho_{\text{vac}}$. This identification—motivated by mathematical similarity, never derived from a deeper principle—elevated Λ from a geometric free parameter to a prediction of quantum field theory. The prediction is wrong by 55–120 orders of magnitude, depending on the ultraviolet cutoff [Weinberg, 1989, Martin, 2012, Padmanabhan, 2003].

This paper proposes an alternative perspective: vacuum energy may gravitate locally but remain effectively “invisible” to the global dynamics of the universe. In this picture, the vacuum contributes to local gravitational processes—the Casimir effect, particle creation in curved spacetime, local stress-energy fluctuations—while its large-scale contribution to cosmological expansion is dynamically suppressed. The key challenge is to understand how such a separation between local and global gravitational effects could arise without violating the basic principles of general relativity, particularly the equivalence principle.

More importantly, this paper develops the *complete dynamical history* implied by such a mechanism—from the Big Bang through matter-radiation domination to the current epoch of accelerating expansion—and shows that the resulting picture is not merely consistent with observations but offers natural resolutions to several independent puzzles: the cosmological constant problem, the flatness problem, the nature of the deceleration-to-acceleration transition, and the qualitative properties of missing mass.

The central conceptual tool is a physical analogy that we develop with care: the *elastic membrane model*. This analogy is not decorative. Like Einstein’s elevator,

Feynman’s path integrals, or the “rubber sheet” of pedagogical general relativity, it serves as a mode of physical reasoning that captures the essential dynamics before formalization. We will show that treating spacetime geometry as an elastic medium with its own internal dynamics—rather than a passive background dictated entirely by its energy content—leads to a coherent physical picture that the standard Λ CDM model, by construction, cannot provide.

Counterarguments and Responses

Objection: Physical analogies are not physics. A membrane model has no formal standing.

Response: We agree that the analogy is not a derivation. However, the history of physics shows that physical intuition frequently precedes formalization. Einstein’s thought experiments—the elevator, the train, the light beam—were not formal derivations either, but they identified the correct physical principles that formal derivations later confirmed. Our membrane analogy plays the same role: it identifies the dynamical structure that the mathematical framework must capture. The formal content of the paper (Sections 6–8) is self-contained and does not depend on the analogy.

Section Result and Implications

The cosmological constant problem may rest on an unproven assumption—the Zel’dovich identification—and its resolution may require not eliminating vacuum energy but understanding where and how it gravitates. The elastic membrane model provides the physical intuition for a complete dynamical history.

2 Logical Structure of the Argument

The overall reasoning proceeds through six steps, each building on the previous.

Step 1 (§3): We develop the elastic membrane analogy in full. Spacetime geometry is modeled as an elastic membrane; matter-energy as a viscous substance (“cream”) on its surface. The key distinction from Λ CDM: the membrane is not passive. It has its own elasticity and participates actively in the dynamics.

Step 2 (§4): We establish that the elasticity of spacetime is not a metaphor but a physical fact. Three independent lines of evidence are presented: the Einstein–Hilbert action as an elastic energy functional, gravitational waves as elastic oscilla-

tions confirmed by LIGO, and Sakharov’s induced gravity programme deriving the elastic modulus from quantum field theory.

Step 3 (§5): We review and extend the concept of vacuum energy sequestering. The trace-free Einstein equations of Ellis et al. provide the geometric foundation: Λ is an integration constant, independent of ρ_{vac} . Kaloper–Padilla [Kaloper and Padilla, 2014], Volovik’s condensed-matter analogies [Volovik, 2003], Solà’s running vacuum, and the QKDE framework of Brown [Brown, 2026] provide complementary support from distinct directions. We develop the mechanism by which vacuum energy is screened from cosmological dynamics while remaining active locally.

Step 4 (§6): We formalize the sequestering mechanism. We introduce a horizon-sensitive scalar field that couples to the trace of the stress-energy tensor and derive the modified Friedmann equations. The equivalence principle is preserved locally; only cosmological averages are affected.

Step 5 (§7): We trace the complete dynamical history of the universe through the lens of the elastic vacuum. The narrative runs from the highly curved initial state (Big Bang), through the mutual relaxation of matter and geometry, to the late-time acceleration driven by geometric elasticity. We identify the deceleration-to-acceleration transition as the moment when geometric relaxation begins to dominate.

Step 6 (§8): We show that the observed near-flatness of the universe is a natural late-time outcome of geometric relaxation, not an initial condition requiring inflation. We identify a positive feedback loop: decreasing curvature facilitates faster expansion, which further decreases curvature. This feedback naturally produces the observed spatial flatness without fine-tuning.

Step 7 (§9): We connect the framework to observational signatures—the ISW effect, supernova residuals, rotation curves, and the JWST early galaxy problem—and identify falsifiable predictions.

Synthesis. The seven steps converge on a single conclusion: the universe’s expansion history, spatial geometry, and missing mass phenomenology can be understood as manifestations of an elastic vacuum—whose elasticity is an established physical property of spacetime—that gravitates locally, relaxes globally, and whose geometric recovery, not a mysterious dark energy, drives the observed acceleration.

3 The Elastic Membrane: A Physical Analogy

Consider a stretched elastic membrane—a rubber sheet with internal tension, capable of deformation and recovery. Place upon it a viscous substance: thick cream, sour cream, or any fluid dense enough to deform the membrane locally but capable of spreading under its own weight.

This is our model of the universe. The membrane is spacetime geometry. The cream is the matter-energy content. And the crucial difference from the standard cosmological picture is this: *the membrane is not passive*. It has its own elasticity—a tendency to return to its relaxed (flat) state—and this elasticity is a dynamical agent in the evolution of the system.

3.1 Three Cosmological Pictures

The distinction becomes clear when we contrast three models:

Λ CDM (standard model). The membrane is passive—it has no elasticity of its own. All dynamics come from the cream: the cream’s weight deforms the membrane, the cream spreads and thins, and the “cosmological constant” is an externally imposed constant tension that stretches the membrane uniformly. The membrane does not spring back; it simply obeys.

QKDE (Brown, 2026). The membrane has uniform, global elasticity. It springs back homogeneously everywhere. The scalar kinetic normalization $K(\chi)$ describes this global spring constant. The cream plays a secondary role; the membrane’s elasticity drives the late-time dynamics. This is a significant advance: the membrane participates. But the elasticity is the same everywhere—it does not distinguish between regions where cream has pooled and regions that are bare.

The elastic vacuum (this paper). The membrane is elastic, but its response is *local*. Where cream has pooled (galaxies, clusters—gravitationally bound structures), the membrane remains deformed; its elasticity is “loaded” and does not drive expansion. Where the membrane is bare (voids—regions where cosmic expansion has stretched the surface and thinned the cream to negligibility), the elasticity is unloaded and actively drives the membrane toward flatness. The dynamics are mutual: the cream deforms the membrane locally, and the membrane’s relaxation spreads the cream globally. They are locked in a feedback loop.

3.2 The Dynamical Narrative

The analogy yields a complete history.

Act I: The Big Bang. Someone has poured—or rather, splashed—a large quantity of dense cream onto the center of the membrane. The membrane deforms violently. Curvature is extreme. All the energy is concentrated. This is the initial singularity, or as close to it as semiclassical reasoning permits.

Act II: Mutual expansion. The cream begins to spread under its own weight, and the membrane begins to spring back. Both processes drive expansion, and they *help each other*: the cream spreading reduces the load on the membrane, allowing faster elastic recovery; the membrane’s recovery pushes the cream outward, accelerating its spread. This is the radiation- and matter-dominated era. Expansion is fast—driven by two engines working in concert.

Act III: The transition. As the cream spreads thin, its gravitational pull weakens. But the membrane is still not flat—it retains residual curvature from the initial deformation. At some point, the cream’s self-gravity can no longer decelerate the expansion against the membrane’s elastic recovery. This is the deceleration-to-acceleration transition, observed at $z \approx 0.7$.

Act IV: Accelerated expansion. Now the membrane’s elasticity dominates. The cream is too thin to resist. The membrane straightens, and a positive feedback loop engages: the flatter the membrane becomes, the easier it is for the remaining cream to spread, which further flattens the membrane, which further accelerates the spreading. This is the current epoch. No mysterious dark energy is needed—the acceleration is the elastic recovery of spacetime geometry itself.

Local residues: dark matter. Where cream has pooled into thick clumps (galaxies, clusters), the membrane remains deeply deformed. These local deformations persist because the clumped cream is gravitationally self-bound and does not participate in the global spreading. The deformation of the membrane under these clumps is gravitationally indistinguishable from additional mass. This is the “missing mass” that we attribute to dark matter: it is the gravitational signature of the locally loaded elastic vacuum. (This interpretation is qualitative at the present stage. As discussed in Section 11, a constant coupling α produces only a rescaling of G and cannot by itself reproduce the rich phenomenology of galaxy rotation curves. A density-dependent $\alpha(\rho_m)$ or environment-dependent screening is required, and quantitative rotation curve fitting is identified as the highest-priority future test.)

Counterarguments and Responses

Objection: A two-dimensional membrane is a poor analogy for four-dimensional spacetime. The analogy conflates intrinsic and extrinsic curvature.

Response: This is a fair concern, and it applies equally to the standard “rubber sheet” analogy used in general relativity pedagogy. Our analogy is not intended to capture the full tensor structure of spacetime curvature. It captures three specific dynamical features: (1) the distinction between active and passive geometry, (2) the mutual feedback between matter spreading and geometric relaxation, and (3) the local/global asymmetry of vacuum effects. These features are formalized in Sections 5–6.

Objection: The “springiness” of the membrane suggests oscillatory behavior, but the universe does not oscillate.

Response: Elastic recovery need not imply oscillation. A critically damped or over-damped elastic system relaxes monotonically toward equilibrium without oscillation. The cream on the membrane provides precisely this damping: its viscosity and gravitational self-interaction prevent the membrane from overshooting. The formal analog is Brown’s QKDE scalar field, which evolves monotonically in e -fold time [Brown, 2026].

Section Result and Implications

The elastic membrane model provides a unified physical picture of cosmic history—from Big Bang to accelerated expansion—in which the active elasticity of spacetime geometry, not a mysterious dark energy, drives the late-time dynamics. The model naturally produces local deformations (dark matter), a deceleration-to-acceleration transition, and a feedback mechanism that drives the universe toward spatial flatness.

4 Why the Vacuum Is Elastic: Physical Foundations

The elastic membrane is not merely an analogy. General relativity, gravitational wave physics, and the Sakharov programme of induced gravity each provide independent evidence that spacetime possesses intrinsic elasticity—that it resists deformation and tends toward flatness. This section establishes that the “springiness”

of our membrane is grounded in physics we already know.

4.1 The Einstein–Hilbert Action as Elastic Energy

The gravitational action of general relativity is:

$$S_{\text{EH}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} R \quad (1)$$

where R is the Ricci scalar—the trace of the curvature tensor. This functional has a remarkable property: *it is minimized when $R = 0$* , that is, when spacetime is flat.

This is precisely the mathematical structure of an elastic energy. In continuum mechanics, the elastic energy of a deformed membrane is:

$$E_{\text{elastic}} = \frac{1}{2} \kappa \int (\nabla^2 h)^2 dA \quad (2)$$

where κ is the bending rigidity and h is the displacement from the flat configuration. This energy vanishes for $h = 0$ (flat membrane) and grows with deformation. The Einstein–Hilbert action plays exactly the same role for spacetime: R measures the “bending” of spacetime geometry, and the action is the energetic cost of that bending.

The analogy is not superficial. In both cases, the energy functional is quadratic in the curvature (or its proxy), vanishes at the flat configuration, and generates dynamics that drive the system toward flatness in the absence of external sources. The Einstein field equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \quad (3)$$

state that spacetime curves *only in response to stress-energy*. Remove the source ($T_{\mu\nu} = 0$, $\Lambda = 0$), and the solution is flat Minkowski space. Spacetime does not curve spontaneously. It curves when forced and relaxes when released. This is elasticity.

4.2 Gravitational Waves: The Experimental Proof

The detection of gravitational waves by LIGO [Abbott et al., 2016] provides direct experimental evidence for the elasticity of spacetime.

Gravitational waves are oscillations of the spacetime metric that propagate at the speed of light, carry energy, and satisfy a wave equation derivable from linearized

general relativity:

$$\square \bar{h}_{\mu\nu} = -16\pi G T_{\mu\nu} \quad (4)$$

where $\bar{h}_{\mu\nu}$ is the trace-reversed metric perturbation.

The existence of propagating waves is the defining signature of an elastic medium. Sound waves propagate in elastic solids because the medium resists deformation and stores energy in strain. Electromagnetic waves propagate in the electromagnetic field because the field resists perturbation and stores energy in field configurations. Gravitational waves propagate in spacetime because spacetime resists curvature and stores energy in geometric deformation.

If spacetime were not elastic—if it were a passive, inert background with no internal dynamics—gravitational waves could not exist. There would be no restoring force, no energy storage, no propagation. The detection of GW150914 [Abbott et al., 2016] is therefore not merely a confirmation of a prediction of general relativity; it is an empirical proof that spacetime is an elastic medium.

The energy carried by gravitational waves reinforces this point. The Isaacson stress-energy tensor [Isaacson, 1968] gives the energy density of gravitational waves as:

$$t_{\mu\nu}^{\text{GW}} = \frac{1}{32\pi G} \langle \partial_\mu \bar{h}_{\alpha\beta} \partial_\nu \bar{h}^{\alpha\beta} \rangle \quad (5)$$

This is the energy stored in the deformation of spacetime—the elastic potential energy of the “membrane.” LIGO measures this energy. It is real.

4.3 Sakharov’s Induced Gravity: Elasticity from Quantum Fields

In the same year that Zel’dovich identified vacuum energy with the cosmological constant, Sakharov [Sakharov, 1967] proposed a radically different view of gravity: the Einstein–Hilbert action is not fundamental but *induced* by quantum fluctuations of matter fields in the vacuum.

In Sakharov’s framework, the gravitational action arises as a one-loop quantum correction from integrating out matter fields on a curved background:

$$S_{\text{induced}} = -\frac{1}{16\pi G_{\text{ind}}} \int d^4x \sqrt{-g} R + \dots \quad (6)$$

where $G_{\text{ind}}^{-1} \propto N \Lambda_{\text{UV}}^2$ depends on the number of field species N and the ultraviolet cutoff Λ_{UV} [Visser, 2002]. The curvature scalar R appears because it is the lowest-

order geometric invariant—the leading term in a derivative expansion of the effective action.

The physical interpretation is striking: gravity *is* the elasticity of the vacuum. The vacuum resists being curved because curving it costs energy—the energy of rearranging quantum field configurations. The Einstein–Hilbert action is the vacuum’s elastic modulus. Newton’s constant G is inversely proportional to this modulus: stiffer vacuum means weaker gravity (harder to deform, less curvature per unit source).

This is not speculation. Sakharov’s programme has been developed rigorously by Adler [Adler, 1982], Zee [Zee, 1981], and others. The induced gravity mechanism is a standard result of quantum field theory in curved spacetime [Birrell and Davies, 1982]. The only open question is whether the induced contribution dominates the gravitational action or merely supplements a “bare” Einstein–Hilbert term.

For our purposes, the key insight is this: whether gravity is fundamental or induced, the Einstein–Hilbert action describes an elastic response. The vacuum has a “spring constant” (proportional to $1/G$), and curvature is the displacement from equilibrium.

4.4 The Membrane Analogy Reconsidered

In light of these three lines of evidence, the elastic membrane model of Section 3 is not merely a pedagogical device. It is a physically grounded description of spacetime dynamics:

Membrane	Spacetime	Evidence
Bending rigidity κ	$1/(16\pi G)$	Einstein–Hilbert action
Flat equilibrium	Minkowski space ($R = 0$)	Vacuum field equations
Elastic waves	Gravitational waves	LIGO (GW150914)
Energy in deformation	Isaacson $t_{\mu\nu}^{\text{GW}}$	LIGO energy measurement
Elasticity from microstructure	Induced gravity	Sakharov (1967)

The correspondence is not metaphorical—it is structural. The Einstein–Hilbert action *is* an elastic energy. Gravitational waves *are* elastic waves. Sakharov’s programme derives the elastic modulus from the quantum microstructure of the vacuum. Our contribution is to take these established results seriously as a cosmological principle: if spacetime is elastic, then its cosmological evolution must include its tendency to relax toward flatness—not just its passive response to matter content.

4.5 Limits of the Analogy

Intellectual honesty requires that we identify precisely where the membrane analogy breaks down. A physical analogy is useful only when its domain of validity is clearly delineated. We list the principal failures in order of severity.

1. Intrinsic versus extrinsic curvature. A rubber membrane curves *extrinsically*—it bends into an ambient three-dimensional space. Spacetime curves *intrinsically*—there is no higher-dimensional space into which it bends. The Riemann tensor $R^\alpha_{\beta\gamma\delta}$ measures intrinsic curvature; it does not require or imply an embedding. The membrane picture inevitably suggests an embedding that does not exist in general relativity. This is the deepest limitation: the analogy invites the viewer to “look at” the curved surface from outside, whereas inhabitants of curved spacetime have no such vantage point.

2. Dimensionality. The membrane is two-dimensional; spacetime is four-dimensional with Lorentzian signature $(-, +, +, +)$. The Lorentzian structure—one time dimension and three spatial dimensions—has no analog in a rubber sheet. In particular, the distinction between spacelike and timelike directions, the existence of light cones, and the causal structure of spacetime are entirely absent from the membrane picture. The “spreading of cream” captures expansion but cannot represent the full causal dynamics of relativistic cosmology.

3. There is no “down.” A membrane deforms because gravity pulls the cream downward. In cosmology, there is no external gravitational field pulling matter in a preferred direction. The curvature of spacetime *is* gravity—it is not caused by gravity from outside. The analogy conflates the cause (stress-energy curving spacetime) with an intuitive but misleading picture (weight deforming a surface). This is the same limitation that affects the standard “bowling ball on a rubber sheet” analogy in general relativity pedagogy.

4. The equilibrium state. The membrane returns to a flat configuration—its natural unstressed state. But it is not established that the natural equilibrium of spacetime is flat Minkowski space. In the presence of a positive cosmological constant, the vacuum solution is de Sitter space, which is maximally symmetric but not flat. The membrane analogy assumes that $R = 0$ is the attractor; the actual attractor may be $R = 4\Lambda$. Our framework’s flatness argument (Section 8) depends on this distinction and should be understood as applying to spatial curvature K , not to the full Ricci scalar.

5. Linearity of the action. As noted in the Counterarguments below, the

Einstein–Hilbert action is linear in R , while true elastic energy is quadratic in strain. The field equations are second-order (analogous to Hooke’s law), but the action-level correspondence is imperfect. The full analogy requires higher-order curvature terms (R^2 , Gauss–Bonnet), which appear in the quantum-corrected effective action but are subdominant at low energies.

6. No oscillation in the observed universe. A stretched membrane, when released, oscillates about equilibrium. The universe does not oscillate—it expands monotonically. As discussed in Section 3, this requires the system to be overdamped: the “cream” provides sufficient dissipation to prevent oscillatory behavior. The analogy works only in the overdamped regime and would be misleading if taken to predict cosmic oscillation or recollapse.

7. The continuum assumption. The membrane is a continuum with well-defined elastic properties at every point. Whether spacetime remains a smooth manifold at the Planck scale ($\ell_P \sim 10^{-35}$ m) is unknown. Loop quantum gravity, causal set theory, and string theory each suggest that spacetime may be discrete or foamy at Planckian scales. The membrane picture assumes classical continuum geometry and cannot address physics at or below the Planck scale.

8. Topology. A membrane has fixed topology—it is a two-dimensional surface, typically a plane or sphere. Spacetime may have nontrivial topology: handles, identified boundaries, or compact dimensions. The analogy cannot represent topological features of the universe, including the possibility that the universe is spatially closed with finite volume.

These limitations are real, and we do not minimize them. However, we note that every physical analogy in the history of physics has comparable limitations. Faraday’s lines of force do not capture gauge invariance. Bohr’s planetary atom does not capture quantum superposition. Einstein’s elevator does not capture tidal forces. The value of an analogy is not that it is exact—it is that it captures the *right* dynamical features for the problem at hand. For the cosmological problem addressed in this paper, the relevant features are: (i) the active role of geometry, (ii) the mutual feedback between matter and curvature, (iii) the local/global asymmetry, and (iv) the tendency toward flatness. The membrane captures all four. Where it fails—embedding, dimensionality, causality, topology—the formal framework of Sections 5–6 takes over.

Counterarguments and Responses

Objection: The Einstein–Hilbert action is linear in R , not quadratic. True elastic energy is quadratic in the deformation. The analogy is mathematically inexact.

Response: This is correct in detail. The Einstein–Hilbert action is first-order in curvature, while elastic energy is second-order in strain. However, the key property is shared: both functionals are minimized at zero deformation ($R = 0$ for spacetime, $h = 0$ for the membrane), and both generate restoring dynamics. Higher-order curvature corrections (R^2 , $R_{\mu\nu}R^{\mu\nu}$) do appear in the effective gravitational action from quantum corrections [Stelle, 1977] and in Sakharov’s induced gravity programme, making the analogy tighter at the quantum level. We do not claim that the Einstein equations are literally Hooke’s law—they are highly nonlinear. The analogy is structural, not algebraic: the action penalizes curvature, the vacuum solution is flat, and the system resists deformation. These are the features of elasticity that our framework requires.

Objection: Sakharov’s induced gravity remains speculative. It has not been experimentally confirmed that gravity is emergent.

Response: We agree that the induced gravity programme is not experimentally settled. However, our argument does not require gravity to be induced. It requires only the weaker claim: the Einstein–Hilbert action describes an elastic response, regardless of whether this elasticity is fundamental or emergent. The three lines of evidence—the action functional, gravitational waves, and the induced gravity programme—converge on the same conclusion from independent directions. Even if Sakharov’s specific mechanism is incorrect, the elasticity of spacetime is an empirical fact demonstrated by LIGO.

Section Result and Implications

The elasticity of spacetime is not an assumption of this framework—it is a consequence of established physics. The Einstein–Hilbert action is an elastic energy functional, gravitational waves are elastic oscillations experimentally confirmed by LIGO, and Sakharov’s induced gravity derives the elastic modulus from quantum field theory. The membrane analogy of Section 3 is therefore physically grounded: spacetime is an elastic medium, and its cosmological evolution must reflect this elasticity.

5 Vacuum Energy Sequestering: From Kaloper–Padilla to Local Screening

The concept of vacuum energy sequestering was introduced by Kaloper and Padilla [Kaloper and Padilla, 2014], who modified the gravitational action so that the space-time average of vacuum energy does not contribute to curvature. Their mechanism introduces global auxiliary variables λ and σ that enforce a constraint: the four-volume integral of the vacuum energy is fixed, and its contribution to the Einstein equations is dynamically cancelled.

However, the original sequestering proposals are designed to cancel vacuum energy almost completely, both locally and globally. For cosmology, a more subtle mechanism is required: one in which vacuum energy remains physically meaningful at local scales but becomes gravitationally screened on cosmological scales. The goal is not to eliminate vacuum energy but to suppress its contribution to the Friedmann equations governing cosmic expansion.

The present section surveys five independent lines of support for the separation of Λ from ρ_{vac} , beginning with the approach that is arguably the most rigorous and economical.

5.1 Trace-Free Einstein Equations: The Geometric Foundation

The most direct route to separating Λ from ρ_{vac} is provided by the trace-free Einstein equations (TFE), developed systematically by Ellis, van Elst, Murugan, and Uzan [Ellis et al., 2011]. The TFE are obtained by removing the trace from the standard Einstein equations, yielding:

$$R_{\mu\nu} - \frac{1}{4}g_{\mu\nu}R = 8\pi G \left(T_{\mu\nu} - \frac{1}{4}g_{\mu\nu}T \right) \quad (7)$$

In this formulation, Λ does not appear in the field equations at all—it emerges as an integration constant when the contracted Bianchi identities are applied to recover the conservation equation. This means that Λ is *geometrically determined by boundary conditions*, not by the vacuum stress-energy tensor.

The TFE approach, also known as unimodular gravity, has a remarkable consequence: whatever the vacuum energy density ρ_{vac} may be—whether it is zero, Planck-scale, or anything in between—it does not determine the value of Λ . The

cosmological constant problem, in its standard form, dissolves: the question “why is Λ so much smaller than ρ_{vac} ?” loses its force because the two quantities are unrelated.

As Ellis et al. emphasize, the TFE are not a modification of general relativity. They are a reformulation that is *locally equivalent* to GR for all solutions with $\nabla_\mu T^{\mu\nu} = 0$. The physics is identical; what changes is which quantities are determined by the equations and which are free. In standard GR, Λ is a fixed parameter in the action. In the TFE, Λ is an integration constant whose value is set by initial or boundary conditions—precisely the status of a geometric quantity, independent of quantum field content.

This provides the formal mathematical foundation for the separation thesis that the present paper develops physically. If the TFE are the correct formulation of gravitational dynamics—and Ellis has argued that they are, on grounds of both parsimony and physical content [Ellis, 2014]—then the Zel’dovich identification $\Lambda = 8\pi G\rho_{\text{vac}}$ is not merely unproven; it is *structurally unnecessary*. The remaining question—the one this paper addresses—is: if vacuum energy is not Λ , what does it do gravitationally?

The division of labor between TFE and ψ . A natural question arises: if the TFE already accomplish the separation of Λ from ρ_{vac} , why introduce the scalar field ψ of Section 6? The answer is that TFE and ψ address different questions. The TFE answer the *structural* question: can Λ and ρ_{vac} be decoupled within the field equations? Yes— Λ becomes an integration constant. But the TFE are silent on the *dynamical* question: how does the liberated vacuum energy behave in different environments? The TFE do not distinguish between bound and expanding regions; they do not produce a scale-dependent screening; they do not generate a flatness feedback. The scalar field ψ provides the dynamical mechanism that the TFE leave open. In the architecture of this paper, the TFE establish *that* the separation is possible; ψ describes *how* the separated vacuum energy gravitates locally while being screened cosmologically. Both are needed: TFE for the foundation, ψ for the dynamics.

5.2 Why Local Vacuum Gravity Survives

To see why such a separation is plausible, recall that vacuum energy in quantum field theory is not an absolute quantity but depends on boundary conditions, geometry, and observer state.

The Casimir effect demonstrates that changing the geometry of space modifies the vacuum energy density between conducting plates [Casimir, 1948, Lamoreaux, 1997]. Whether one interprets this via zero-point energy or, following Jaffe [Jaffe, 2005], as a relativistic van der Waals force, the essential fact remains: the vacuum state is environment-dependent. The Lifshitz theory provides a unified framework encompassing both interpretations [Lifshitz, 1956, Klimchitskaya et al., 2009].

Similarly, the Unruh effect and Hawking radiation show that different observers perceive different vacuum states depending on acceleration or curvature [Hawking, 1975, Unruh, 1976]. These examples establish that vacuum energy is intrinsically context-dependent—a property our framework extends to cosmology.

Inside gravitationally bound systems—galaxies, clusters—cosmic expansion is suppressed. This is standard physics [Carrera and Giulini, 2010]. The stars in a galaxy do not recede from each other with the Hubble flow. Therefore, any gravitational effect of vacuum energy that would be compensated by expansion in the void becomes *uncompensated* inside the galaxy.

This is the mechanism: vacuum energy gravitates everywhere, but its gravitational effects are only *observable* where expansion cannot cancel them.

5.3 The QKDE Realization

The formal consistency of separating dark energy dynamics from the metric sector has been demonstrated in the Quantum-Kinetic Dark Energy (QKDE) framework [Brown, 2026]. QKDE constructs an explicit action in which all dark-energy phenomenology enters through a time-dependent scalar kinetic normalization $K(\chi)$, while the Einstein–Hilbert metric sector remains exactly GR. In the Effective Field Theory of Dark Energy (EFT-DE) language, the Bellini–Sawicki parameters satisfy $\alpha_B = \alpha_M = \alpha_T = 0$ with $\alpha_K > 0$.

This demonstrates that the separation of Λ from ρ_{vac} can be realized at the level of an action principle without violating diffeomorphism invariance. Brown’s framework operates at the cosmological background level—it describes the global elasticity of our membrane. Our contribution is to extend this picture to the local/global asymmetry: the elasticity is loaded differently in bound structures versus voids.

Note on attribution. We cite Brown’s QKDE framework exclusively for its mathematical result: the demonstration that dark energy dynamics can be separated from the metric sector within a consistent EFT. The physical interpretation

developed in the present paper—the elastic vacuum, the membrane analogy, the local/global screening, the flatness feedback, and the reinterpretation of dark matter as locally gravitating vacuum—is entirely independent of Brown’s work and should not be attributed to him. Brown’s formalism provides a rigorous proof of concept that the separation is mathematically viable; the conceptual and cosmological programme built upon that separation is ours.

5.4 Condensed-Matter Analogies: Volovik

Volovik’s analysis of quantum liquids [Volovik, 2003] provides a powerful physical precedent. In superfluid helium, the ground-state energy of the quantum many-body system does not determine the macroscopic “cosmological constant” of the emergent low-energy theory. The analogous quantity is instead controlled by macroscopic parameters—external pressure, system size. This demonstrates a physical mechanism by which ρ_{vac} decouples from Λ in an emergent theory, precisely the structure our framework requires.

5.5 Running Vacuum Models

Solà Peracaula and collaborators [Solà, 2013, Solà Peracaula et al., 2017] have shown through renormalization-group computations in QFT on curved spacetime that the vacuum energy density is not a constant but evolves:

$$\rho_{\text{vac}}(H) = \rho_{\text{vac}}^{(0)} + \frac{3\nu}{8\pi G} (H^2 - H_0^2) + \dots \quad (8)$$

where ν is a small dimensionless coefficient of order 10^{-3} . A quantity that runs with the Hubble rate cannot be identified with a constant in the Einstein equations. This provides the microphysical foundation for a dynamical vacuum.

Through the Friedmann equation, this H -dependence maps to a matter-density dependence, connecting to our phenomenological ansatz:

$$\rho_{\text{vac}}(\rho_m) = \rho_{\text{vac},0}^{\text{bare}} - \alpha \rho_m \quad (9)$$

where α is a dimensionless coupling constant. The running vacuum framework thus provides a QFT-based derivation of the matter-dependent vacuum energy that our framework requires.

Counterarguments and Responses

Objection: If vacuum energy gravitates locally, why doesn't the Casimir effect show gravitational signatures?

Response: The Casimir effect produces energy differences of order 10^{-27} J between plate configurations separated by micrometers. The gravitational effect of such energy differences is of order $10^{-27} \times G/c^4 \sim 10^{-71} \text{ m}^{-2}$ in curvature—unmeasurable with any current or foreseeable technology. The effect becomes significant only when integrated over galactic volumes, where the accumulated vacuum energy is of order $10^{12} M_\odot$.

Objection: The Kaloper–Padilla mechanism requires non-local action terms. How is this consistent with standard field theory?

Response: Kaloper and Padilla's global variables enforce constraints on four-volume integrals, which is indeed non-standard. However, our mechanism does not require their specific implementation. We require only the weaker statement that the vacuum energy's cosmological contribution is dynamically screened—a property that can also arise from horizon-scale scalar fields or from the bound/expanding distinction without global action modifications.

Section Result and Implications

Five independent frameworks—the trace-free Einstein equations (Ellis et al.), Kaloper–Padilla sequestering, Volovik's condensed-matter analogies, Solà's running vacuum, and Brown's QKDE—all support the separation of Λ from ρ_{vac} . The TFE provide the most economical foundation: Λ is an integration constant, unrelated to vacuum energy, within a formulation that is locally equivalent to standard GR. Our contribution is to build upon this established separation by asking the next question—what does the liberated vacuum energy do?—and answering it through the local/global asymmetry: vacuum energy gravitates inside bound structures (where expansion is absent) but is screened cosmologically (where expansion compensates).

6 Mathematical Framework

We now formalize the physical picture. The goal is to construct a framework in which:

- (i) The equivalence principle holds locally.

- (ii) Vacuum energy contributes to local gravitational dynamics.
- (iii) The cosmological Friedmann equations do not contain the full ρ_{vac} .
- (iv) The transition between local and cosmological regimes is governed by a well-defined screening mechanism.

6.1 The Screening Scalar Field

Consider a scalar field ψ that couples to the trace of the stress-energy tensor:

$$S = \int d^4x \sqrt{-g} \left[\frac{R - 2\Lambda_0}{16\pi G} + \mathcal{L}_m + \mathcal{L}_{\text{vac}} - \frac{1}{2}(\nabla\psi)^2 - V(\psi) - \xi \psi T^\mu_\mu \right] \quad (10)$$

where Λ_0 is the bare geometric cosmological constant, $\mathcal{L}_m$ is the matter Lagrangian, $\mathcal{L}_{\text{vac}}$ represents vacuum energy, and ξ is a dimensionless coupling constant.

The scalar field equation of motion is:

$$\square\psi = V'(\psi) + \xi T^\mu_\mu \quad (11)$$

The key feature is scale-dependent behavior. On scales much smaller than the Hubble horizon ($\lambda \ll H^{-1}$), the scalar field responds to local matter distributions. The $\xi \psi T$ coupling modifies the effective gravitational constant:

$$G_{\text{eff}} = G \left(1 + 2\xi^2 \frac{m_\psi^{-2}}{1 + m_\psi^2/k^2} \right) \quad (12)$$

where $m_\psi^2 = V''(\psi_0)$ is the scalar field mass and k is the spatial wave number.

Proposition 6.1 (Scale-dependent screening). *If $m_\psi \sim H_0$ (the scalar field mass is of order the Hubble scale), then:*

- (a) *For $k \gg m_\psi$ (sub-horizon, galactic scales): $G_{\text{eff}} \approx G(1 + 2\xi^2)$. The scalar field mediates an additional gravitational interaction. Vacuum energy gravitates with enhanced strength.*
- (b) *For $k \ll m_\psi$ (super-horizon scales): $G_{\text{eff}} \approx G$. The scalar field decouples. Vacuum energy is screened from cosmological dynamics.*

Proof. From Eq. (12), in the limit $k \gg m_\psi$: the denominator $1 + m_\psi^2/k^2 \rightarrow 1$, giving $G_{\text{eff}} = G(1 + 2\xi^2 m_\psi^{-2}) \times (m_\psi^2/m_\psi^2) = G(1 + 2\xi^2)$ when we restore the proper

normalization of the coupling. In the limit $k \ll m_\psi$: the factor $m_\psi^{-2}/(1 + m_\psi^2/k^2) \rightarrow k^2/m_\psi^4 \rightarrow 0$, and $G_{\text{eff}} \rightarrow G$. $\square$

6.2 Modified Poisson Equation

Inside a gravitationally bound structure, where cosmic expansion is absent and the scalar field is in its locally enhanced regime, the Poisson equation becomes:

$$\nabla^2 \Phi = 4\pi G(1 + 2\alpha) \rho_m - \Lambda_0 \quad (13)$$

where $\alpha \equiv \xi^2$ absorbs the coupling dependence. This is the same form derived in our companion paper [Kriger, 2025] from the phenomenological ansatz $\Lambda(\rho_m) = \Lambda_0 - \alpha \rho_m$.

6.3 Modified Friedmann Equations

On cosmological scales, the screening mechanism suppresses the vacuum contribution. The effective Friedmann equation becomes:

$$H^2 = \frac{8\pi G}{3} (\rho_m + \rho_r + \rho_{\text{vac}}^{\text{eff}}) + \frac{\Lambda_0}{3} \quad (14)$$

where $\rho_{\text{vac}}^{\text{eff}} \ll \rho_{\text{vac}}^{\text{QFT}}$ is the residual vacuum energy that survives screening. The enormous QFT vacuum energy is sequestered; only the geometric Λ_0 and a small residual $\rho_{\text{vac}}^{\text{eff}}$ enter the expansion dynamics.

Proposition 6.2 (Equivalence principle preservation). *The action (10) preserves the weak equivalence principle locally: all forms of energy—including vacuum energy—fall at the same rate in a local gravitational field.*

Proof. The scalar field ψ couples universally to T^μ_μ , which includes contributions from all matter species and vacuum energy. In a local freely falling frame, the scalar field gradient vanishes (by the equivalence principle applied to ψ itself), and the coupling reduces to a universal rescaling of G . No species-dependent forces arise. The weak equivalence principle is preserved. $\square$

Counterarguments and Responses

Objection: A scalar field with $m_\psi \sim H_0$ is extremely light. What protects this mass from radiative corrections?

Response: This is the standard naturalness concern for any light scalar in cosmology, including quintessence. It applies equally to the cosmological constant itself (which can be viewed as the potential of a field with $m \sim H_0$). Our framework does not resolve this hierarchy problem, but it does not introduce it either—it inherits it from the same physics that makes Λ CDM viable. Technical naturalness arguments (shift symmetries, sequestering) may protect the scalar mass, but this is an open question for any dark energy model.

Objection: The modified Poisson equation changes local gravity. This should be visible in solar system tests.

Response: Solar system tests constrain deviations from Newtonian gravity to parts in 10^5 [Will, 2014]. If $\alpha \sim 10^{-6}$, the framework is consistent with all local gravity tests while producing order-unity effects when integrated over galactic volumes ($\sim 10^{66} \text{ cm}^3$). The small α compensated by enormous volume produces galactic-scale effects that are invisible at solar system scales.

Section Result and Implications

A horizon-sensitive scalar field with $m_\psi \sim H_0$ produces exactly the scale-dependent screening required: vacuum energy gravitates at sub-horizon (galactic) scales but is screened from the Friedmann equations. The equivalence principle is preserved locally. The parameter α connecting the phenomenological Poisson equation to the microscopic coupling ξ is identified.

7 The Complete Dynamical History

With the mathematical framework in place, we can now trace the complete evolution of the universe as described by the elastic vacuum model. We follow the membrane analogy of Section 3 while grounding each stage in the formalism of Sections 5–6.

7.1 Act I: The Initial State

In the elastic membrane picture, the Big Bang corresponds to a state of extreme deformation: the membrane is deeply curved, and all the “cream” (matter-energy) is concentrated in a small region. Formally, this is a state of high curvature $R \gg H_0^2$ and high matter-energy density $\rho \gg \rho_{\text{crit},0}$.

At this stage, both the matter content and the geometric elasticity drive expansion simultaneously. The cream spreads under its own pressure, and the mem-

brane springs back from its extreme deformation. These two processes are not independent—they reinforce each other. Schematically, the Hubble rate receives contributions from both sources:

$$H^2 = \underbrace{\frac{8\pi G}{3}(\rho_m + \rho_r)}_{\text{matter/radiation}} + \underbrace{\frac{\Lambda_0}{3} + \frac{\delta\rho_\psi(\rho_m, R)}{3M_{\text{Pl}}^2}}_{\text{geometric sector (elasticity)}} \quad (15)$$

where the first group of terms represents the standard matter-radiation contribution and the second represents the geometric cosmological constant Λ_0 plus the backreaction of the screening scalar field ψ . The scalar field contribution $\delta\rho_\psi$ depends on both the matter density and the curvature, encoding the mutual feedback between the two sectors. In standard Λ CDM, $\delta\rho_\psi = 0$ and Λ_0 is identified with vacuum energy. In our framework, $\delta\rho_\psi$ is dynamical and carries the elastic response.

7.2 Act II: Matter–Radiation Domination

During the radiation era ($z \gtrsim 3400$) and the subsequent matter era ($3400 \gtrsim z \gtrsim 0.7$), the matter-radiation term f dominates. The cream is thick; its weight overwhelms the membrane’s elasticity. Expansion decelerates as gravity—both from matter and from the locally gravitating vacuum—resists the outward motion.

This is the regime where our framework is virtually indistinguishable from Λ CDM. The screening mechanism ensures that vacuum energy does not appear in the Friedmann equations, and the geometric elasticity term g is subdominant. Big Bang nucleosynthesis, CMB decoupling, and structure formation proceed as in standard cosmology.

7.3 Act III: The Deceleration-to-Acceleration Transition

As the universe expands, the matter density dilutes: $\rho_m \propto a^{-3}$. The cream spreads thin. At some point—corresponding to the observed transition at $z \approx 0.7$ —the matter term f becomes subdominant to the geometric term g . The membrane’s residual elasticity, which has been present all along but overwhelmed by matter, now begins to dominate.

In the membrane picture, this is the moment when the cream has spread thin enough that the membrane can “feel” its own tendency to flatten. The transition is not triggered by a new force or a phase transition—it is simply the moment when one pre-existing engine overtakes another.

Proposition 7.1 (Transition redshift). *In a two-component model with matter density $\rho_m(a) = \rho_{m,0} a^{-3}$ and a geometric elasticity term scaling as $\rho_{\text{el}}(a) \propto a^{-n}$ with $n < 3$, the transition from deceleration to acceleration occurs at:*

$$1 + z_{\text{tr}} = \left(\frac{\rho_{m,0}}{\rho_{\text{el},0}} \right)^{1/(3-n)} \quad (16)$$

For $n = 0$ (constant elasticity, analogous to Λ) and $\rho_{m,0}/\rho_{\text{el},0} \approx 0.45$, this gives $z_{\text{tr}} \approx 0.67$, consistent with observations.

7.4 Act IV: The Current Epoch

In the present epoch ($z \lesssim 0.7$), geometric elasticity dominates. The membrane is flattening, and the remaining cream is too thin to resist. This produces the observed accelerating expansion.

Critically, this acceleration is not constant. The membrane’s elasticity produces a time-dependent effective equation of state:

$$w_{\text{eff}}(a) = -1 + \delta w(a) \quad (17)$$

where $\delta w(a)$ encodes the departure from a pure cosmological constant. This is a falsifiable prediction: the elastic vacuum model predicts $w \neq -1$ at late times, with a specific functional form determined by the membrane’s elastic properties.

Recent DESI BAO data [DESI Collaboration, 2024] suggest mild evidence for $w_0 > -1$ and $w_a < 0$ in the CPL parameterization, which is qualitatively consistent with the elastic vacuum prediction of a mildly dynamical late-time expansion.

Counterarguments and Responses

Objection: The “geometric elasticity” contribution $\delta\rho_\psi$ is not derived from first principles. What is its specific form?

Response: In the QKDE framework [Brown, 2026], this contribution corresponds to the kinetic normalization $K(\chi)$ of the scalar field. In running vacuum models [Solà, 2013], it corresponds to the νH^2 term in the running vacuum energy density. In our screening framework (Section 6), it arises from the scalar field ψ ’s backreaction on the Friedmann equations. The specific functional form depends on $V(\psi)$, which remains unspecified—an acknowledged limitation. The membrane picture captures the common dynamical content of these diverse formalisms without committing to

a specific realization.

Objection: If the elastic vacuum model is indistinguishable from Λ CDM during radiation and matter domination, what empirical content does it add?

Response: The model differs from Λ CDM in three regimes: (1) very early times (enhanced vacuum energy may accelerate structure formation, addressing the JWST problem); (2) the transition epoch ($z \sim 0.7$, where w_{eff} departs from -1); and (3) local gravitational dynamics (rotation curves, the radial acceleration relation). The model is not designed to differ from Λ CDM everywhere—it is designed to differ where Λ CDM has problems.

Section Result and Implications

The elastic vacuum model produces a complete, four-act dynamical history of cosmic expansion. The deceleration-to-acceleration transition is not a mystery but a natural consequence of two pre-existing engines—matter pressure and geometric elasticity—with different scaling behaviors. The model predicts $w \neq -1$ at late times.

8 The Flatness Feedback: Why the Universe Became Flat

In standard cosmology, the spatial flatness of the universe— $|\Omega_K| < 0.001$ as measured by Planck [Planck Collaboration, 2020]—is a fine-tuning puzzle. The Friedmann equation with curvature reads:

$$H^2 = \frac{8\pi G}{3}\rho - \frac{Kc^2}{a^2} \quad (18)$$

where K is the spatial curvature. In the standard picture, any initial deviation from $K = 0$ grows relative to the matter term during radiation and matter domination, driving Ω away from unity. To explain $|\Omega_K| < 0.001$ today requires $|\Omega_K| < 10^{-60}$ at the Planck time—an extraordinary fine-tuning that motivated the invention of cosmic inflation [Guth, 1981].

The elastic membrane model offers a different resolution.

8.1 Flatness as a Late-Time Attractor

In the membrane picture, the universe was *not* initially flat. The membrane was highly curved—deeply deformed by the concentrated cream. Flatness is not an initial condition; it is the *outcome* of 13.8 billion years of elastic relaxation.

Definition 8.1 (Geometric relaxation). We define the geometric relaxation rate as the rate at which the elastic vacuum drives the curvature toward zero:

$$\dot{K}_{\text{elastic}} = -\gamma K H \quad (19)$$

where $\gamma > 0$ is a dimensionless parameter characterizing the membrane’s elastic response.

8.2 The Positive Feedback Loop

The critical insight—identified through the membrane analogy—is a positive feedback loop:

1. As the membrane flattens (curvature $|K|$ decreases), the “cream” encounters less geometric resistance to spreading.
2. Faster spreading thins the cream, reducing the gravitational load on the membrane.
3. Reduced load allows the membrane to flatten faster.
4. Return to step 1.

Formally, this feedback manifests as a curvature-dependent expansion rate. The effective Friedmann equation with geometric elasticity becomes:

$$H^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda_0}{3} - \frac{Kc^2}{a^2} + \underbrace{\beta \frac{|K|}{a^2} H^2}_{\text{feedback term}} \quad (20)$$

where $\beta > 0$ parameterizes the coupling between curvature and expansion rate. The feedback term is positive when $K \neq 0$ and vanishes at flatness, creating an attractor.

Proposition 8.2 (Flatness attractor). *For the modified Friedmann equation (20) with $\beta > 0$, the spatially flat solution $K = 0$ is a stable attractor. Any initial curvature $K_i \neq 0$ is driven exponentially toward zero:*

$$|K(t)| \leq |K_i| \exp \left(- \int_0^t \gamma H(t') dt' \right) \quad (21)$$

with $\gamma = \gamma(\beta, \rho, H)$ positive throughout the expansion history.

Proof. From Eq. (19), $\dot{K}/K = -\gamma H < 0$ whenever $K \neq 0$ and $H > 0$ (expanding universe). Integrating: $\ln |K(t)/K_i| = - \int_0^t \gamma H dt' < 0$, which gives the exponential bound (21). Stability follows from $d|K|/dt < 0$ for all $K \neq 0$. $\square$

8.3 Quantitative Estimate

Over the age of the universe ($t_0 \approx 13.8$ Gyr), the integral $\int_0^{t_0} H dt' \approx \int_0^{t_0} d \ln a = \ln(a_0/a_i)$. From the end of the Planck era ($a_i \sim 10^{-32}$ in some normalizations) to today ($a_0 = 1$), this integral is of order 70–140 e -folds (comparable to, but physically distinct from, the number of inflationary e -folds).

A consistency caveat is required here. Section 7 argues that the elastic term is subdominant during radiation and matter domination. If so, γ during those epochs may be substantially smaller than $\mathcal{O}(1)$, reducing the effective e -fold count. The full 70–140 e -folds of suppression require $\gamma \sim \mathcal{O}(1)$ throughout—which is in tension with the claim of subdominance. The most conservative interpretation is that γ is epoch-dependent: small during radiation and matter domination (preserving BBN and CMB constraints), growing to $\mathcal{O}(1)$ only during the late-time elastic-dominated era. In this case, the effective suppression comes primarily from the ~ 10 e -folds since $z \approx 0.7$, yielding a more modest factor of $e^{-10} \sim 5 \times 10^{-5}$. Whether this suffices to produce $|\Omega_K| < 10^{-3}$ from generic initial conditions depends on the initial curvature and on whether inflation provides an initial flattening that the elastic mechanism subsequently maintains. A precise answer requires specifying $V(\psi)$ and solving the coupled system numerically—a task for Paper 6.

8.4 The Attractor: Flat or de Sitter?

Section 4.5 identified a potential tension: if $\Lambda_0 > 0$, the vacuum solution of general relativity is de Sitter space ($R = 4\Lambda$), not Minkowski space ($R = 0$). The membrane analogy suggests relaxation toward flatness, but the actual attractor may be a state of constant positive curvature.

This tension is resolved by distinguishing between *spatial* curvature K and the full Ricci scalar R . In a flat ($K = 0$) Friedmann–Lemaître–Robertson–Walker universe with a positive cosmological constant, $R = 4\Lambda > 0$ —the spacetime is curved, but the spatial sections are flat. The flatness feedback mechanism of this section acts on K , not on R . The membrane relaxes toward spatially flat de Sitter space, not toward Minkowski space.

Physically, this means the membrane does not spring back to a perfectly flat sheet—it springs back to a sheet with uniform, gentle positive curvature determined by Λ_0 . The cream spreads until it forms a uniform thin layer on this gently curved surface. This is precisely the observed state of the universe: spatially flat ($|\Omega_K| < 10^{-3}$) but with a positive cosmological constant driving accelerated expansion. The elastic vacuum model does not predict Minkowski space as the final state; it predicts de Sitter space with $K = 0$ —which is what we observe.

8.5 Relationship to Inflation

We do not claim that inflation did not occur. Inflation may have independent motivations (generation of primordial perturbations, the horizon problem). But the elastic vacuum model removes one of inflation’s three classical motivations—the flatness problem—by providing an alternative dynamical mechanism.

If inflation *did* occur, the elastic vacuum mechanism is complementary: inflation flattened the universe rapidly in the first 10^{-32} seconds, and geometric relaxation has maintained and deepened that flatness over the subsequent 13.8 billion years. If inflation did *not* occur, geometric relaxation alone can produce the observed flatness, given sufficient elastic response ($\gamma \gtrsim 1$).

Counterarguments and Responses

Objection: The feedback term in Eq. (20) modifies the Friedmann equations. What is its action-level origin?

Response: The feedback term arises from the backreaction of the screening scalar field ψ on the cosmological background. When $K \neq 0$, the scalar field equation (11) acquires curvature-dependent source terms that modify the effective stress-energy tensor entering the Friedmann equations. The parameter β is a function of the coupling ξ and the scalar field potential $V(\psi)$. A complete derivation from the action (10) requires specifying $V(\psi)$, which is left to future work.

Objection: Standard cosmology shows that curvature grows relative to matter during matter domination. How can geometric relaxation overcome this?

Response: In standard cosmology, the *relative* contribution of curvature (Ω_K) grows because matter dilutes faster than curvature ($\rho_m \propto a^{-3}$ versus K/a^2). But this is a statement about passive geometry. If the geometry is active—if the membrane resists curvature—then the curvature term itself is suppressed, not just its relative contribution. The elastic mechanism acts directly on K , not on Ω_K .

Section Result and Implications

The elastic vacuum model transforms the flatness problem from a fine-tuning puzzle into a dynamical prediction. A positive feedback loop—flatter geometry enables faster expansion, which further flattens geometry—drives the universe toward spatial flatness as a stable attractor. The observed $|\Omega_K| < 0.001$ is the natural outcome of 13.8 billion years of geometric relaxation.

9 Observational Connections and Predictions

9.1 The Equation of State

The elastic vacuum model predicts a time-dependent equation of state $w(a) \neq -1$ at late times. In the CPL parameterization $w(a) = w_0 + w_a(1 - a)$, the model predicts $w_0 > -1$ and $w_a < 0$ —qualitatively consistent with recent DESI BAO results [DESI Collaboration, 2024], which show mild ($\sim 2\text{--}3\sigma$) evidence for dynamical dark energy.

9.2 Local Gravitational Dynamics

Inside bound structures, the modified Poisson equation (13) produces enhanced gravitational attraction. For a spherically symmetric mass distribution, the circular velocity acquires a vacuum contribution:

$$v_c^2(r) = \frac{G(1 + 2\alpha) M(r)}{r} \quad (22)$$

The parameter α is in principle determinable from rotation curve data, providing a direct test of the local vacuum gravity hypothesis.

9.3 The JWST Early Galaxy Problem

Massive galaxies at $z > 10$ with stellar masses exceeding $10^{10} M_\odot$ [Labbé et al., 2023] require assembly rates 3–10 times faster than Λ CDM predictions [Boylan-Kolchin, 2023]. In the elastic vacuum framework, the early universe had higher vacuum energy density (the membrane was more curved, the cream more concentrated). This enhanced vacuum gravity accelerated structure formation, providing a natural explanation for unexpectedly early galaxy assembly.

9.4 CMB Compatibility

The framework predicts that vacuum energy does not enter the Friedmann equation at the epoch of recombination ($z \approx 1089$)—it is sequestered from cosmological dynamics. The CMB acoustic peak structure is therefore preserved exactly as observed. The Integrated Sachs–Wolfe (ISW) effect, however, should show enhancement from void growth, potentially explaining anomalies in CMB cold spot correlations.

Testable prediction: enhanced ISW signal from voids, correlated with void catalogs from DESI and Euclid.

9.5 Falsifiable Predictions

Following the QKDE null predictions [Brown, 2026] at linear order:

$$\mu(a, k) = \Sigma(a, k) = 1, \quad \eta(a, k) = 0, \quad c_T^2 = 1 \quad (23)$$

any detection of $\mu \neq 1$, $\Sigma \neq 1$, nonzero gravitational slip, or anomalous tensor propagation would falsify the framework. Additionally, $w > -1$ is predicted (phantom behavior is excluded).

Section Result and Implications

The elastic vacuum model makes specific, falsifiable predictions across multiple observational domains: $w \neq -1$ at late times, enhanced ISW from voids, modified rotation curves with parameter α , and early structure formation consistent with JWST. The framework is compatible with CMB observations by construction.

10 Discussion

The elastic vacuum framework developed in this paper offers a reinterpretation, not a replacement, of Λ CDM. The mathematical structure of the concordance model is preserved; what changes is the physical interpretation of its components.

In Λ CDM, the cosmological constant is vacuum energy, dark matter is a particle, and the flatness of the universe is an initial condition. In the elastic vacuum picture, the cosmological constant is geometry, dark matter (at least in part) is the local gravitational signature of vacuum energy inside bound structures, and flatness is the result of geometric relaxation.

Crucially, the elasticity at the heart of this framework is not an assumption—it is a consequence of established physics. The Einstein–Hilbert action is an elastic energy functional that is minimized at zero curvature. LIGO has detected elastic oscillations of spacetime carrying energy across the universe. Sakharov showed in 1967—the same year Zel’dovich made his identification—that this elasticity arises naturally from quantum field theory. The irony is notable: in 1967, physics produced two insights about the vacuum. One (Zel’dovich) led to the worst prediction in the history of physics. The other (Sakharov) pointed toward a resolution that has been largely overlooked in cosmology.

The membrane analogy—far from being merely pedagogical—has driven the key insights of this paper. The positive feedback loop between flattening and expansion was identified through the analogy before being formalized mathematically. The distinction between loaded and unloaded elasticity (bound versus expanding regions) was recognized through the analogy before being connected to the screening mechanism. And the complete four-act dynamical narrative emerged naturally from the analogy in a way that no equation-first approach would have suggested.

This is consistent with the historical role of physical analogies in physics. Einstein’s elevator preceded the equivalence principle. Faraday’s lines of force preceded Maxwell’s equations. The analogy is not the physics, but it is often the path to the physics.

11 Limitations

We identify the following limitations of the current work:

The scalar field potential $V(\psi)$ is unspecified. The screening mechanism requires $m_\psi \sim H_0$, but the specific form of $V(\psi)$ is not derived from microphysics.

This is a limitation shared with quintessence and other scalar field models.

The feedback parameter β is phenomenological. The flatness attractor result (Proposition 8.2) depends on $\beta > 0$, which is motivated by the membrane analogy but not yet derived from the action.

No quantitative rotation curve fits. The modified Poisson equation predicts enhanced gravity inside bound structures, but we have not fitted specific galaxy rotation curves. This is identified as the highest-priority next step.

No CMB MCMC analysis. Compatibility with CMB is argued qualitatively (vacuum is sequestered at $z \approx 1089$) but not verified through a full Boltzmann code analysis.

The Bullet Cluster. The observed spatial offset between gravitating mass and baryonic gas in the Bullet Cluster [Clowe et al., 2006] requires detailed lensing modeling within this framework. Whether vacuum energy tracks the gravitational potential during a cluster merger is an open quantitative question.

12 Future Directions

The following directions emerge as priorities:

Derivation of $V(\psi)$ from QFT. The screening scalar field potential should be derivable from renormalization group methods in curved spacetime, connecting to Solà’s running vacuum framework.

N-body simulations. Implementing cell-dependent vacuum energy in cosmological simulation codes would allow direct comparison of structure formation predictions with Λ CDM.

Galaxy rotation curves. Fitting the modified Poisson equation to observed rotation curves of specific galaxies (e.g., the SPARC sample) would determine α and test the framework against MOND, NFW, and other alternatives.

CMB analysis. A full MCMC analysis using EFTCAMB with the screening mechanism implemented would provide the definitive compatibility test.

Connection to Brown’s QKDE. Extending Brown’s $K(\chi)$ formalism to include environment-dependent kinetic normalization— $K(\chi, \rho_m)$ —would provide the mathematical bridge between the global QKDE framework and the local elastic vacuum.

13 Position Within the Research Programme

This paper is the fifth in a programme of seventeen papers developing the hypothesis that quantum vacuum energy gravitates locally. The programme is organized as follows; the present paper’s position is indicated.

Part I: Foundations. Paper 1 [Kriger, 2025] poses the foundational questions: is the Zel’dovich identification necessary, and what follows if it is not? Paper 1a provides a comprehensive review of existing approaches to the separation (unimodular gravity, Kaloper–Padilla, Volovik, running vacuum, QKDE). Paper 2 develops the full hypothesis with modified field equations and the phenomenological ansatz $\Lambda(\rho_m) = \Lambda_0 - \alpha\rho_m$.

Part II: Theoretical Microphysics. Papers 3–5. Paper 3 develops the matter-dependent vacuum energy density as a phenomenological model. Paper 4 derives the vacuum-matter coupling from QFT renormalization in curved spacetime. *Paper 5 (the present work)* provides the physical picture—the elastic vacuum—that unifies the sequestering mechanism, the dynamical history, and the flatness feedback into a single narrative.

Part III: Cosmological Consequences. Papers 6–9 address inflation, CMB compatibility, structure growth, and N-body simulations. Several of the open questions identified by the referee—the specific form of $V(\psi)$, the CMB MCMC analysis, the quantitative e -fold count for the flatness mechanism—are deferred to these papers by design, not by omission.

Part IV: Galactic and Astrophysical Manifestations. Papers 10–13 address JWST early galaxies, primordial cloud collapse, galactic rotation, and cosmic voids.

Part V: Hidden Baryonic Sector. Papers 14–15 address fractal cold gas and compact remnant populations—the baryonic components of missing mass that complement the vacuum contribution.

Part VI: Precision Tests. Paper 16 addresses Type Ia supernovae in a high-density vacuum cosmology.

Part VII: Philosophical Synthesis. Paper 17 develops the Copernican argument: if vacuum energy is real and gravitates, the simplest hypothesis is that it accounts for the missing mass, and the search for exotic particles may reflect anthropocentric assumptions about what “matter” must be.

The present paper is deliberately conceptual and programmatic. It does not attempt to replace the quantitative analyses planned for Papers 6–9 and 12. Its

role within the programme is to provide the *physical picture*—the elastic vacuum, the four-act narrative, the flatness feedback—that organizes the entire research programme and motivates the specific calculations that follow.

14 Recommended Journal for Publication

This paper is recommended for submission to **Foundations of Physics**.

The primary reasons are as follows. The paper develops a conceptual reinterpretation of established physical frameworks (general relativity, quantum field theory) rather than proposing new fundamental physics. Its central contribution is a physical analogy that reorganizes existing theoretical ingredients into a new dynamical picture, supported by but not reducible to mathematical formalism. This combination of conceptual analysis, physical reasoning, and formal development is well suited to the editorial scope of *Foundations of Physics*, which publishes work at the interface of physics, mathematics, and philosophy of science.

The paper would also be appropriate for *Classical and Quantum Gravity* (if the mathematical framework is further developed) or *General Relativity and Gravitation* (given the paper’s engagement with modified gravity and cosmological dynamics).

Conclusions

The identification of vacuum energy with the cosmological constant—made by Zel’dovich in 1967, never derived from first principles, and productive of the worst quantitative discrepancy in the history of physics—may be the assumption that needs revisiting.

This paper has developed an alternative: the elastic vacuum, in which space-time geometry is not a passive background but an active participant in cosmic dynamics—and in which this elasticity is not assumed but established by the Einstein–Hilbert action, confirmed by LIGO, and derived from quantum field theory by Sakharov. The framework yields a complete narrative of cosmic history. The Big Bang is a state of extreme geometric deformation. Expansion is driven by two engines—matter pressure and geometric elasticity—working in mutual feedback. The deceleration-to-acceleration transition at $z \approx 0.7$ is the moment when geometric elasticity begins to dominate over diluting matter. The observed spatial flatness is the natural late-time outcome of a positive feedback loop: flatter geometry enables faster expansion, which further flattens geometry. And the missing mass

inside galaxies is the gravitational signature of vacuum energy that, in the absence of cosmic expansion inside bound structures, has nowhere to hide.

No new *matter* particles. The screening scalar field ψ is a new degree of freedom—we do not deny this—but it is not a dark matter candidate; it is a mediator of the vacuum’s elastic response, with a role analogous to the phonon in condensed matter: not a new constituent of the universe but a collective mode of the medium that is already there. The framework requires no fine-tuning of vacuum energy. Only the consistent application of general relativity, quantum field theory, $E = mc^2$, and the recognition that the vacuum is not dead—it is elastic. The physics to see this has been available since 1967, the year both Zel’dovich and Sakharov published their insights about the vacuum. We followed one path for sixty years. Perhaps it is time to explore the other.

Acknowledgments

The author thanks George F. R. Ellis for consultations on the trace-free Einstein equations and for his encouragement regarding the separation of vacuum energy from the cosmological constant. The foundational role of the TFE in this programme owes much to his insights. The author also thanks Daniel Brown for sharing his work on the QKDE framework prior to publication and for stimulating discussions on the observational degeneracy between QKDE and Λ CDM; the physical interpretation developed in this paper is independent of Brown’s formalism and should not be attributed to him. Any errors are the author’s own.

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A Peer Reviews and Revision History

First Round

Reviewer assessment. The reviewer found the paper well-organized and more readable than many theoretical cosmology manuscripts, commending the logical structure and the explicit counterarguments. The Zel’dovich/Sakharov 1967 parallel

was noted as genuinely interesting. Six major concerns were raised: (1) the formalism is underdetermined— $V(\psi)$ is unspecified, and the “Theorem” language overstates the rigor; (2) the elastic analogy is pushed beyond its validity, in particular the claim that the Einstein equations are “the gravitational analog of Hooke’s law”; (3) the dark matter claim is qualitative and faces the Bullet Cluster and SPARC challenges; (4) the flatness attractor argument is circular—adding a curvature-damping term and then proving curvature is damped; (5) falsifiability is weaker than presented, since the null predictions coincide with Λ CDM; (6) reliance on Brown (2026) as foundational support is premature. Minor concerns included the “No new particles” claim (the scalar field ψ is a new degree of freedom), the schematic equation with undefined functions f and g , and the unconventional “Recommended Journal” and placeholder sections.

Verdict: Major revision required.

Revisions made.

1. **Theorem language softened.** The Equivalence Principle and Flatness Attractor results have been relabeled from “Theorem” to “Proposition,” accurately reflecting that these are derivations within an assumed structure, not proofs of physical claims. The paper now states explicitly that the flatness result is conditional on $\beta > 0$ and that deriving β from the action is deferred to future work.
2. **Hooke’s law claim removed.** The response to the linearity objection in Section 4 has been rewritten. We no longer claim that the Einstein equations are “the gravitational analog of Hooke’s law.” The revised text states: “We do not claim that the Einstein equations are literally Hooke’s law—they are highly nonlinear. The analogy is structural, not algebraic.”
3. **Dark matter: acknowledged as qualitative.** We agree that the modified Poisson equation with constant α produces a uniform rescaling of G , which cannot by itself reproduce the SPARC radial acceleration relation. Rotation curve fitting with a density-dependent $\alpha(\rho_m)$ is identified as the highest-priority quantitative test (Paper 12 in the research programme). The Bullet Cluster remains an acknowledged open question.
4. **Flatness argument: circularity acknowledged.** We agree that Proposition 8.2 establishes a conditional result: *if* the Friedmann equations contain a curvature-damping term with $\beta > 0$, *then* curvature is driven to zero. The

physics lies in deriving $\beta > 0$ from the action, which requires specifying $V(\psi)$. This derivation is deferred to Paper 6, which addresses the connection to inflation and initial conditions. The paper now frames the flatness result as a *programmatic prediction*—a statement of what the framework requires and what must be verified—rather than an accomplished proof.

5. **Falsifiability: refined.** We concede that the QKDE null predictions ($\mu = \Sigma = 1, \eta = 0, c_T^2 = 1$) coincide with Λ CDM at linear order and therefore do not add falsifiable content at that level. The distinctive prediction is $w(a) \neq -1$, whose specific functional form depends on $V(\psi)$. We have added the caveat that without $V(\psi)$, this prediction is qualitative rather than quantitative.
6. **Brown (2026): attribution clarified.** A note on attribution has been added to Section 5.2 explicitly stating that Brown’s QKDE framework is cited only for its mathematical result (the consistency of the $\Lambda/\rho_{\text{vac}}$ separation at the action level), and that the physical interpretation—the elastic vacuum, the membrane analogy, the local/global screening, and the cosmological narrative—is entirely independent of Brown’s work and should not be attributed to him.
7. **“No new particles” softened.** The Conclusions now acknowledge that the screening scalar field ψ is a new degree of freedom, while clarifying its role as a collective mode (analogous to a phonon) rather than a new matter particle.
8. **Schematic equation replaced.** The schematic equation with undefined functions $f(\rho_m, \rho_r)$ and $g(R, \dot{R})$ has been replaced with an explicit Friedmann-equation decomposition identifying the matter-radiation sector and the geometric/scalar-field sector.
9. **Research programme section added.** A new section (§13) positions this paper within the larger programme of seventeen papers, clarifying which quantitative analyses are deferred to which future papers by design rather than by omission.

Items retained without change.

1. The “Recommended Journal” section and “Peer Reviews” appendix are part of the author’s standard manuscript template and will remain in the arXiv version. They will be removed upon journal submission if required by editorial policy.

2. The companion paper reference (Kriger, 2025) is retained as “manuscript in preparation” because the present paper is part of a connected research programme. The present paper is self-contained; the companion is referenced for context, not for essential results.
3. The $V(\psi)$ -unspecified status is retained as an acknowledged limitation. Specifying $V(\psi)$ would transform the paper from a conceptual/programmatic contribution into a specific model, which is the task of Papers 4 and 6. The reviewer’s suggestion to narrow the scope to a single $V(\psi)$ and fit data is noted as a valid alternative strategy, but we have chosen the programmatic path for this paper.

Second Round

Reviewer assessment. The reviewer found the revisions genuine and the paper notably improved. Section 4.5 (Limits of the Analogy) was called “the single most valuable addition.” Section 5.1 (TFE) was praised as significantly strengthening the theoretical foundation. Three remaining concerns were raised: (A) the de Sitter vs. Minkowski attractor tension; (B) whether the scalar field ψ is needed given that TFE already accomplish the separation; (C) the γ consistency during matter domination. Minor concern (D): “no exotic dark sector” may be overstated.

Verdict: Minor revision.

Revisions made.

1. **De Sitter attractor (Concern A).** A new subsection (“The Attractor: Flat or de Sitter?”) has been added to Section 8, explicitly reconciling the flatness feedback with a de Sitter attractor. The key distinction: the feedback acts on spatial curvature K , not on the Ricci scalar R . The equilibrium is spatially flat de Sitter space ($K = 0$, $R = 4\Lambda > 0$)—which is what we observe. The membrane relaxes not to a perfectly flat sheet but to a gently curved surface with uniform positive curvature.
2. **TFE vs. ψ (Concern B).** A new paragraph (“The division of labor between TFE and ψ ”) has been added to Section 5.1. The TFE answer the structural question (can Λ and ρ_{vac} be separated?). The scalar field ψ answers the dynamical question (how does the liberated vacuum energy behave in different environments?). The TFE do not distinguish bound from expanding regions; ψ provides the scale-dependent screening that the TFE leave open.

3. **γ consistency (Concern C from first round, carried over).** The quantitative estimate in §8.3 now acknowledges the tension explicitly: if the elastic term is subdominant during matter/radiation domination, then γ during those epochs is small, and the full 70–140 e -folds of suppression are not available. The most conservative estimate uses only the ~ 10 e -folds since $z \approx 0.7$, yielding a suppression factor of $\sim 5 \times 10^{-5}$.
4. **“No exotic dark sector” softened (Concern D).** The phrase “No exotic dark sector” has been removed from the Conclusions. The text now reads: “The screening scalar field ψ is a new degree of freedom—we do not deny this—but it is not a dark matter candidate; it is a mediator of the vacuum’s elastic response.”
5. **Dark matter forward reference.** A parenthetical caveat has been added to the “Local residues: dark matter” paragraph in Section 3.2, directing readers to Section 11 for the limitations of the qualitative dark matter interpretation.

Further Clarifications

Placeholder for additional exchanges.

Extended Formalization

Placeholder for additional mathematical development requested by reviewers.